

ActInf Textbook Group ~ Cohort 3 ~ Meeting 19 (Chapter 8, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_3 | Cohort 3 ~ Meeting 19 (Chapter 8, part 2) | 56:27

All right, it's July 18th, 2023, and we're in our second discussion on Chapter 8.

So, where would anyone like to begin or any general remark on Chapter 8?

Well, I can say something.

I added a couple of questions into the list at the very end on Box 8.1, but we can go through some other stuff first, if you want.

Sounds good.

All right, let's look at the first question.

Thank you.

Perhaps if I put some background in of my understanding.

So, my understanding is we have - well, I think if we go to - there's Raphael Boyatch or Bogatch's paper, a tutorial on free energy, which starts off with looking at a single neuron with an input distribution of a particular mean and variance.

Yeah.

And he then builds that up to the point where you've got a multi-layer system with multiple nodes.

And the synapses represent means and variances.

So, I think there's - we're not seriously suggesting that synapses actually represent particular means and variances or whatever.

But I think there's the idea that collectively a population of neurons is going to encode mean values and variances.

And precision is inverse variance, of course.

So, I suppose my other frame of reference is Gaussian mixture models where you've got Gaussians characterized by weights or amplitude, mean and variance.

So, then going on to box 8.1, I'm confused that there seems to be a way of treating precision as a magnitude, as an amplitude.

Or changing that, changing the Gaussian perhaps by reducing the variance, increasing the precision is going to increase the peak to make it more prominent, which seems a strange way of doing things.

Yeah, exactly.

So, for me, well, precision and attention are not synonyms.

So, attention seems to be something at the neuromodulation timescale as opposed to synaptic weights at a much slower timescale or general influence with usual synaptic - well, some neurotransmitters.

So, yeah, it doesn't make sense how a lot of that box.

You're on mute, Daniel.

Yeah.

Ali or anyone, do you want to give a thought?

Or...

Okay.

So, here, the precision being described is the precision to the sensory input.

And so, equivalent to attention to sensory input in the sense that if the... and reflected by the A matrix.

So, if the incoming sensory data are inferred to be accurate, then they will update the hidden state more.

So, if the sensory data incoming, if the sensory data incoming were not believed to be accurate, low precision, low attention, then observations wouldn't update hidden states.

So, I don't think that that necessarily refers to the psychological concept of attention.

But is your question about...

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weighting rescales the attention to different variables is what dictates the influence that they have on the overall free energy gradient descent. If you have two variables and they both are equally attended to, then the free energy is a 50-50 compromise such that there's going to be a flat manifold where one of them is getting better by one point and one's getting worse by one point. Whereas if you're attending to one of them 10 times more than the other, so it's 10 times more precision on that belief, then each increment of free energy minimization, you'd be willing to accept 10 decrements on the other one. So the attention blends

the role of each parameter being optimized on the overall scalar gradient descent.

Yeah, I've still got a bit of a block on that. I'll go away and have a think about that. But I think what you've said just leads naturally on to my other point. So as I see it, as you say, that rebalancing of the attention. So if we have attention given to extraceptive input, then you're sensing the outside world, you're treating that as important. If you then shift to proprioceptive attention, then that's more likely to lead or that would lead to action. Or there's the whole thing about sensory attenuation. Why does sensory attenuation happen? Is it a random thing?

Well, there's the splitting across and there's a neuroscientific model framework. And then there's also actual, we're trying to do some engineering stuff here. But on the neuroscience side, well, generally, are we saying that there is some oscillation,

flip flopping between sense and effectively action? Or is it just something that's happening randomly that's that things will drift into an action phase or a sensory balance?

Yeah, I think there's a lot of pieces there. So systems don't have to have sensory attenuation.

If having a constant regime of attention

is simply how the system is defined by fiat, or at some level of generative modeling for some variable, there may not be any reason to propose this cognitive phenomena. However, whether you look at eye circling and visual input, where we have like precision, high precision on the sensory input when we're fixed, and then we pull off the gas when we move so that we suppress input while the visual field is very shifted, and then re-engage. So whether we think about like the eye circade attention to visual

input, or if we think about the motor control example in box 8.1, which is exactly analogous, it's just interoceptive instead of exteroceptive, where there's a suppression of sensory input input to support action selection instead of belief updating. In both those cases, a kind of adaptive, emergent, cybernetic pattern is this oscillation of attention to enable the benefits of high attention to sensory input, input, but then also be able to transiently engage in other mode where input to sensory data is lowered. So I think that just turns out to be a solution or an approach to navigate the trade-offs, because you wouldn't want to just have high attention or only low attention. So then the navigation solution oscillates between them. So this is, so we're saying that this is just something that happens to be found in real grains or as fit into a model of real grains. So I think that's a good question, I think, of course, because of the way that we're going to think about sensory attenuation, it's not something that appears in the discrete time domain, is it? I guess you could model sensory attenuation in discrete time, but correct, sensory attenuation is not a necessary part of active inference agents. So why is this useful? Because this pattern helps us understand the cognitive similarities between speaking and listening, where the bird song example is a famous case of sensory attenuation. Like you expect the song to be playing, so as long as it's playing, you listen, and then when it goes away, then now you take action. And then while acting, your precision decreases until you eventually cease acting again. So turn taking conversation, eye saccades, motor control with proprioceptive, the cognitive patterns can be identified. And those are accounts of complex phenomena that are first principles and don't appeal to reward or reinforcement. They appeal to some specific computational phenomena that maps between the neurobiology loosely, like you brought up with the mean and the variance, and the statistical parameterization. And then that model generates unique explanations and predictions and say, oh, well, maybe this dyskinesia is a challenge of initiating movement. And this one's a challenge of terminating movements. Yeah. Yes. And there's there's a time with various mental illness as well, isn't there? So schizophrenia and tiding with sensory attenuation. It kind of reminds me of the the work by Adam Saffron of Albus. Because Friston and Carhart Harris had this rebus paper, Relaxed Beliefs Under Psychedelics. And then there was like this other proposals like SEBIS, Strengthened Beliefs Under Psychedelics. And then Adam was like, well, it's kind of just altered. I know that's really general, but it's just like increasing the precision on this one might then decrease the precision on something else. So there's not just a precision variable. It has to be a little bit more unpacked

with precision on what beliefs.

Because we could talk about beliefs on sense states, about beliefs on latent states, beliefs on action.

It's all belief updating. It's all Bayesian beliefs.

Can you think, is there any reference that I perhaps I should read that might get me thinking about precision in a better way?

Maybe the birdsong,

Friston and Frith birdsong example.

And also maybe with Sanji, if we can ask and see, his simulation seemed very amenable to it.

Yeah, I think possibly because I've got it fixed in my mind, the idea of synapses encoding variance.

Whereas, as you say, variance is, there's variance much more broadly.

I think that's where I don't really have a good handle.

I think this is a barrier.

I have a barrier.

So how do I get through the barrier?

Yeah, that's kind of a hard but good topic.

And

it just reminds me of the end of seven
um
here.

Like we call D the prior.

But of course, once you escape the most trivial pedagogical generative model, then it makes more sense to be clear.

This is a prior on hidden states.

Because you're also talking about a prior or hyper prior on other variables.

And also, that was like kind of the motivation for the generalized notation notation.

So that we could, which was originally called generalized subscript notation, because we wanted to be able to talk about like

a preference on preference or a policy on preference or a prior on preference.

Because when you're in the nested models, you have to clarify which term you're talking about.

Hmm.

Okay.

Yeah.

So I'm, I'm pretty much happy with the rest of the chapter.

It's that, it's just that one box that I need to get around.

So yeah.

So like you said, the bird song thing.

So I've read that in the past.

That, that makes sense.

But it's this ambiguity I have around.

The word precision.

And how it.

Well, that, that, that one phrase of precision and attention being synonyms.

It's that that I've got the block on.

Let me try one, one more angle on this.

So we're sitting up and sitting down example from the sensory attenuation.

So let's just say that our proprioceptors give us like 90% confident information.

Um.

When we're in that state, but when we're moving, it gives us.

50, 50.

So it's totally ambiguous.

But then it's 90, 90, 10.

Once we're in the seated or the standing.

Um.

State.

So.

We have a true.

Um.

Hidden state, which is our actual posture.

And then.

Um.

Um.

Let's just say that we're getting seated input.

Because we're seated.

But because of a cognitive decision.

Now we prefer standing input.

Um.

Now if the pragmatic value were scaled super high.

Then you would take the policy to stand up.

Because the pragmatic value would just overwhelm everything else.

However, if the pragmatic value is not scaled super high.

Then there's a little bit of like a frustration.

Because.

Um.

It's your preferred state.

But not so preferred.

That it's overwhelming.

The belief updating.

To be anchored closer to the sitting down.

So you have kind of these two peaks.

You could be getting.

You could be, you know, drawing from this peak of observations sitting down.

Or you could be drawing from the other peak.

Um.

And then.

Um.

Even if this.

Even if the other peak is slightly higher.

The optimization still.

Holds to the left side.

And then like by increasing the variance.

Reduce your precision.

Then.

The.

In the.

In that ambiguity.

From the sensory input.

Then.

The.

The.

The preferred.

State.

Then it.

You can select action.

To stand up.

And kind of.

It's like temporarily flattening the landscape.

So that you can make a move.

Without needing to.

Um.

Want it more.

More.

Yeah.

I think.

Something you said at the beginning.

That was.

Very.

Significant.

I think.

Because I.

I'm thinking.

When we're talking about.

Precision.

It's precision of.

Beliefs.

Of sensory input.

Whereas.

There's also.

Precision on.

In states.

And.

Yeah.

Between priors.

And.

And new input.

Uh.

It's that balance.

See what else.

Happens.

Okay.

Lock of Altera.

Thomas.

Part.

Explains.

Some of this.

Like the.

The historiosity.

Of this.

Lock of Altera.

Was used.

To.

Describe.

Structures.

Controlled.

Sequential.

Activities.

In the.

Continuous.

Time.

Setting.

Okay.

Learning.

A functional.

S.

That plays.

The role.

Of a free.

Energy.

This could definitely be unpacked a little more.

It seems like this is a continuous time.

Free energy based method.

Uh.

For gradients.

Based optimization of mean and variance of the Laplace approximation.

but it seems like it could use a little bit more context

I'm all right in thinking so if we visualize a normal distribution and we're saying we increase the precision we can think of that as the shape the change of shape of the distribution so that the peak goes up so it's it is that that's the right way of thinking about it we're changing the shape of the normal distribution so so hence increasing increasing the position so reducing the variance is is the same thing as increasing the the peak at the at the mean value which is the mode as well yeah yeah

yes it sharpens it when it's more attended to and this is sort of like the two um the two faces of the KL divergence is mean seeking versus mode seeking and attention modulation

um influences which one of those it goes to as well as where that optimization starts in the landscape

because if it's too attentive if it's too attentive to get out of a local optima then it becomes functionally mode seeking

because whichever mode it gets to it'll just think okay I'm doing as well as I can
because also the plus doesn't um the Gaussian has support everywhere but a Laplace is a downward facing parabola
so it doesn't have infinite support whereas so it's not exactly a Gaussian mixture model
um
and then the mode seeking um

it just tries to find
whatever the highest point is and then fit a Gaussian around it
versus the mean seeking tries to cover as much of the distribution
and match the center of gravity
with the central tendency
so in the case of the sitting up and standing standing um
like
you you're you're um usually wanting to be pretty precise about your body's position
so you're mode seeking you're mode seeking you're locally you're sitting down
then
it's like the attention relaxes that moves the KL to be more like mean seeking
now it's like 50 50 50 percent your prior half the time you're standing half you're seated
and the sensory data are ambiguous
and you prefer to be standing
so then that preference breaks the symmetry
and then you can resharpen your attention back when you're standing
so then the KL can have kind of this like end-to-end
um reduction of divergence
and the only knob that you had to change
was the attention
now how to adaptively tune attention
that's a question
but that's why it's it's um interesting conversion to evolution
that we see the same like cyclic attention strategy
in the saccade and in the um standing example
so we see that as a repeated
heuristic
and
umdp chapter 7 has largely superseded the use of generalized lockable terra and active inference
that's kind of interesting
because that's like saying the discrete time has kind of taken over from the continuous time

yeah and i've made the point before so a lot of those
sort of classic examples when including the birdsong in the spm 12
um is is based on the spm adm so active dynamic expectation maximization model
so that's that's all within the continuous
with this continuous time frame uh concept as opposed to pomdp um
so it's it's like that's yeah yeah it has been superseded
uh because all of the com so much of the conversation now is about pomdp
um yeah eventually we'll we'll have every model annotated in the literature so we can look at like the trends
and the um
which ones were used but suffice to say
lockable terra i don't see this coming up recently
and continuous times are definitely not as frequent as pomdp's
i mean pi mdp only describes continuous time
and it's just so much closer to the setting of uh discrete time
explicit planning
counterfactuals uh reinforcement learning reward like
the discrete time formalism is a lot closer to it
yeah i suppose yes more more useful for sort of high level sort of cognitive functioning as opposed to the
basic movement in an environment
but that's kind of the cool like internet of things question like what if you could have a
you know a wing um or like a water you know something
and then all that it was really like endowed with
was heuristics around attention modulation
maybe that only takes like two lines or something
and then like with this heuristic around attention
then it can do like a in a continuous time um
sensory capacity whereas you could you could train a pomdp
and do all these big like matrix calculations i mean that that's sort of the question
where do these um different models have different
computational advantages or explainability advantages
okay
yeah okay thanks yeah oh yeah these are great questions
also the uh
the recent model stream nine
i'm just reloading it with aswin paul
um here it is
he presents uh computational the only other time we've seen computational complexity estimates was with
uh branching time active inference

i thought this was super fascinating so here's um
so classical act inf
si sophisticated inference
time step 30 horizon or two horizon
dp dynamic programming
um aif
one is um
is just single time step act inf
so 30 time step sophistication
10 to the 68
complexity
cognitive complexity
and just like how i just found it really interesting how
sophisticated inference blow up is so high
and and uh
he's using dynamic programming based methods
to learn smooth
priors on preference
so i think there's a lot of different
different
there's a lot of different little motifs
that might have
very effective performance in a certain setting
like in the setting of a camera
that moves
eye circadian
and blinking
by
converge to being an adaptive strategy
so finding out which
generative models and phenomena
parsimoniously describe these different natural systems
might be very helpful for
them being able to compose
um them together
and understand okay where do we need
thank you ali
talk to you later

where do we need different um

different

models

i think i asked him like can you do like

the full can you do like

this approximation and this part of the nested model and that one and this part

and he said yes

so you could even implement two different approximations

for different parts of the same generative model i think

okay

then generalize synchrony

okay here's

where the bird song comes back into play

bird hears the song there's no need to generate it

if the bird predicts a song that is not heard it must start singing

learning here

it's uh described more in the paper i assume that it is learning according to this box

this leads to turn taking sometimes phrase to sing from the same hymn sheet

um

yes uh it it is

and

therefore it's not really like a model of improvisational communication

or necessarily even communication as information transmission

they're literally just it's like there's something that has to be carried and they're just like

one of them carries it until the other one picks up

so it's like it's a very basal

aspect of communication because the model is really just trying to describe turn taking

acting to generate birdsong requires a reduction the precision of predictions

about the consequences of action i'm not sure why that is

why couldn't you take action and also increase your precision about the consequences of your actions

that'd kind of be like action um i mean might even be connected to something like ocd

where action that that um action that when you take it blurs your precision

is kind of like self-terminated

because eventually you'll just wobble out of that behavioral attractor

whereas if there's a behavior

that is that is increasing your precision

so you're attending more and more to the sensory outcomes

that that action is seeking

then i feel like that action becomes

like tighter and tighter um like an attractor

but something like like um i mean even though a lot of times different uh neuropathologies and stuff come up

i think the reality is a lot of these are really sophisticated

multi-level cognitive phenomena and so

it's just kind of pointing in that direction but it's not really it's not it's not the same thing

um

generalized synchrony

um

again sing from the same hymn sheet

they have learned to share the same narrative i think

if you consider the song to be a narrative

and also open question is like how much is it is already like baked in and then it's being fine-tuned

versus it's not like they're starting with two different singing concepts and then actually learning the whole thing but

just

okay then

uh

hybrid models

uh

hybrid models

uh

single models that contain both continuous and discrete variables

in this case the lower level is continuous time

x x' x'' the higher level is um

discrete time past present future

this is the same family of model that

ryan smith et al explored in live stream 46

uh active inference does not contradict folk psychology

where the lower level is the motor active inference and then the higher levels the decision-making active inference

there's supposed to be an edge here

um

um

uh

message passing

uh

uh

uh

uh

procedures

uh

procedures exist

for graphs that have discrete and continuous time

continuous time so they're kind of brought up separately just because of uh pedagogy

but

ultimately it seems like there's no issue with mixing

the uh

uh

the mapping between discrete and continuous time

happens by basically giving discrete

state spaces uh

integer values in some continuous space state space

um

like

um

like

here

you

you could associate the coordinates

with

with the discrete location but then also have it be a continuous space

so then you i guess can have the discrete time model and the continuous model in the same

state space u_m being like kind of described in one from the discrete side as zero one and then in the continuous one from continuous space space from zero to one

[illegible]

zero zero zero zero zero zero zero zero zero zero zero

Probably this is a useful paper to look more into.

16 neural populations.

Four locations and four time steps.

One, two, three, and four.

But I don't know why they stay active. I don't know why they don't drop back down.

Well, and here's what's happening. This is the person is being told to look at the cross.

Then a dot is being shown at the top of the screen.

There's some 200 millisecond delay that represents like visual processing.

Then they accelerate.

Here's their point of maximum speed.

They accelerate and then decelerate and fixate on the cube.

Where the cube was.

And each of these four traces represents beliefs.

But again, I'm not sure why.

Why it's not just like one comes up and then it goes down. The second comes up, it goes down.

Okay, mixture models, Gaussian mixture models like you brought up earlier.

What do you think the relevance is of the Gaussian mixture here?

Neil.

Hmm.

I don't know.

Because there's reference to some delta functions here.

So precise.

Well, totally precise.

Priors there.

So I'm not sure.

Where the Gaussian comes in other than that bottom equation.

The normal distribution.

Okay.

The priors on the mean and the variance are precise.

So we're basically just looking for a point estimate on mean and a point estimate on variance.

Not looking to have a secondary uncertainty about mean and a secondary uncertainty about variance.

So that's the delta part.

And then I guess the L subscript is the number of mixtures.

And maybe different observations are being allocated or learnt to the different mixtures categories.

So the optimal, you know, if you had two, if you had one mixture, they'd be the same as a Gaussian.

If you have three mixtures, then you'd be trying to parameter three different gaussians with different means

and variances.

And then assigning each data point to one of those mixtures.

And then the question of which.

Which number of mixtures of groups should you have?

Is approached through Bayesian model reduction Bayesian information criterion.

That sounds from 3":

That sounds from 3":

That sounds from 3":

That sounds from 3":

That sounds from 3":

That sounds from 3":

That sounds from 3":

That sounds from 3":

That sounds from 3":

That sounds from 3":

Yeah, interesting and kind of very specific.

Sparse and brief chapter.

I feel like it is very short and just barely touches on a few pieces.

But I guess provides information where there's other continuous time models.

What do you think?

Yeah, I thought it was a very good chapter.

This is more where my interest lies, as you might have guessed on the continuous time side of things.

I just have that one difficulty around the box.

Having accepted what's in the box, everything else follows.

Fine.

It all hangs together quite neatly.

But yeah, there are various things that have been said.

I need to go away and digest.

Just like you say, I'm a point estimate on variance.

Whereas I'm thinking of perhaps I'm confusing, getting estimates and variance confused.

Right.

Because the variance included in the estimate.

Right.

And we, variance is like the mean on our variance estimator.

And usually we don't go two levels deep.

The standard deviation, there's not an SD on the SD usually.

That's just the point estimate on mean and variance.

But then it's like, but the variance of the mean, isn't that just the variance?

How is the variance different than the precision on the mean?

Yeah.

I mean, let's think about the class.

You know, the people are 10 feet tall plus or minus five.

So variance is five by virtue of the actual variation in height.

But the precision on the mean, if we sampled a million creatures, might be very low.

So we might have a very confident estimate.

It's 10 plus or minus five.

Or we might have a very imprecise estimate that it's 10 plus or minus one.

So that's the variance.

And then in the variational gradient descent scheme, attention reweights all the variables so that the gradient descent strikes a balance.

But that just begets the question.

How does the attention get trained?

So that in a multivariate optimization, variables are being traded off appropriately, including situations where there's a dynamic reallocation of optimization focus.

So that's the question.

Yeah.

How is how is the whole system?

Well, how are those parameters of the the chaotic attractor tuned so that it does?

It does follow contained trajectories stable rather than just going off and disappearing down, getting stuck into into in one attractor rather than another.

Yeah.

Yeah.

Yeah.

Well, for for a system to be able to be to descend into an attractor and to escape, it kind of has to be placed on the edge of chaos.

Yeah.

Or have some dynamic oscillatory component.

Otherwise, it just gets attracted to where it's attracted to.

But okay, cool.

Thank you.

Well, tomorrow's the first math learning group or or so.

So people are welcome to come to discord if if they want, but otherwise, see you all next time.

That's great.

Thanks, bye.

Bye.

Bye.